Hiroaki Murayama

Curriculum Vitae

✉ hiromura319@gmail.com
My Website LinkedIn ORCID GitHub Revised: June 15, 2026

Summary

I am a final-year medical student with a research background of infectious disease modelling, seeking doctoral training in quantitative infectious disease epidemiology and mathematical modelling. My research centres on the mathematical formulation of infectious disease dynamics, drawing upon mathematical, statistical and computational methods to understand natural history, transmission processes and the impact of control measures across multiple scales. My research to date has involved emerging and re-emerging infections, including mpox (clade I and IIb) and COVID-19, and has included peer-reviewed modelling studies, governmental analyses and public-facing science communication. Through a PhD, I aim to disentangle the mechanisms shaping epidemic patterns in complex and heterogeneous populations, and address key public health questions.

Education

2027 expected **M.D., Department of Medicine, School of Medicine**, *International University of Health and Welfare*, Chiba, Japan
Expected March 2027.

Appointments and Experience

2026 **Academic Visitor**, *Saw Swee Hock School of Public Health, National University of Singapore*, Singapore
Supervisor: Assistant Professor Akira Endo.
- Derived offspring and final-size distributions for mpox clade IIb transmission in a depleted post-2022 sexual contact network to quantify extinction probabilities, persistence of low-level transmission and the risk of major resurgence.
- Presented ongoing work on mpox clade IIb at the CERM (Centre for Epidemic Research and Modelling) research round.

2026 **Clinical Elective Student**, *Division of Dermatology, National University Hospital*, Singapore
Supervisor: Associate Professor Nisha Suyien Chandran.
- Completed a four-week clinical elective in dermatology.
- Gained clinical exposure to outpatient consultations, ward rounds and dermatological procedures.

2020–2024 **Member**, *National COVID-19 Cluster Task Force, Ministry of Health, Labour and Welfare*, Tokyo, Japan
Supervisor: Professor Hiroshi Nishiura.
- Conducted ad-hoc analyses to support risk assessment of the COVID-19 epidemic and to evaluate the impact of public health responses.
- Conducted analyses on Alpha-variant dynamics, vaccine effectiveness against death using population-level data and the impact of healthcare burden on temporal case fatality risk.

2022–2023 **Research Assistant**, *Graduate School of Public Policy, The University of Tokyo*, Tokyo, Japan
Supervisor: Associate Professor Taisuke Nakata.
- Collaborated with economists on research into COVID-19 response strategies.

2021–2023 **Member**, *CoV-Navi*, Japan
- Reviewed scientific evidence on COVID-19 vaccines for public-facing science communication.

2021–2022 **Research Assistant**, *Graduate School of Social Sciences, Chiba University*, Chiba, Japan
Supervisor: Assistant Professor Shouto Yonekura.
- Developed a Bayesian framework for estimating waning vaccine effectiveness from population-level surveillance data in the presence of multiple circulating variants, in collaboration with Dr Akira Endo.

Publication List

Peer-Reviewed Original Research

J12 **Murayama H**†, Asakura TR†, Dickens BL, Boyle D, Foo JH, Jin S, Mukadi PK, Ejima K, Jung S-m, Nishi A, Prem K, Wakamba AM, Saila-Ngita D, Niyukuri D, Endo A. *Role of community and sexual contacts as drivers of MPXV clade I.* Nature Health. 2026 Apr 6.

J11 Jin S, Asakura TR, **Murayama H**, Jung S, Niyukuri D, Nyandwi J, Nkengurutse L, Kamatari O, Lim JT, Endo A, Dickens BSL. *Disentangling temporal trends of clade Ib monkeypox virus transmission in Burundi.* The Journal of Infectious Diseases. 2025 Sep 10;jiaf475.

J10 Jin S, Asakura TR, **Murayama H**, Niyukuri D, Saila-Ngita D, Lim JT, Endo A, Dickens BSL. *Vaccination strategies to achieve outbreak control for MPXV clade I with a one-time mass campaign in sub-Saharan Africa: a scenario-based modelling study.* PLOS Medicine. 2025 Sep 5;22(9):e1004726.

J9 Ejima K†, Wang Y†, Endo A†, **Murayama H**, Goh YS, Cook AR, Jeong YD, Iwami S, Park H, Dickens BSL, Jin S, Lim JT, Chan CEZ, Chia PY, Young BE, Chio M, Lye DC, Ajelli M. *Evaluating the effectiveness of international travel controls to identify MPXV-infected travellers: a simulation study.* BMC Medicine. 2025 Aug 12;23(1):473.

J8 Asakura TR, Jung S, **Murayama H**, Ghaznavi C, Sakamoto H, Teshima A, Miura F, Endo A. *Modelling international spread of clade IIb mpox on the Asia continent.* Bulletin of the World Health Organization. 2025;103(7):429–436.

J7 Jung S†, Miura F†, **Murayama H**, Funk S, Wallinga J, Lessler J, Endo A. *Dynamic landscape of mpox importation risks driven by heavy-tailed sexual contact networks among men who have sex with men in 2022: a mathematical modelling study.* The Journal of Infectious Diseases. 2024;jiae433.

J6 **Murayama H**, Pearson CAB, Abbott S, Miura F, Jung S, Fearon E, Funk S, Endo A. *Accumulation of immunity in heavy-tailed sexual contact networks shapes mpox outbreak sizes.* The Journal of Infectious Diseases. 2024 Jan 15;229(1):59–63.

J5 **Murayama H**, Endo A, Yonekura S. *Estimation of waning vaccine effectiveness from population-level surveillance data in multi-variant epidemics.* Epidemics. 2023;100726.

J4 Endo A, **Murayama H**, Abbott S, Ratnayake R, Pearson CAB, Edmunds WJ, Fearon E†, Funk S†. *Heavy-tailed sexual contact networks and monkeypox epidemiology in the global outbreak, 2022.* Science. 2022 Sep 25;0(0):eadd4507.

J3 Ko Y, **Murayama H**, Yamasaki L, Kinoshita R, Suzuki M, Nishiura H. *Age-dependent effects of COVID-19 vaccination and healthcare burden on COVID-19 deaths in Tokyo, Japan.* Emerging Infectious Diseases. 2022;28(9).

J2 Yamasaki L†, **Murayama H**†, Hashizume M. *The impact of temperature on the transmissibility and virulence of COVID-19 in Tokyo, Japan.* Scientific Reports. 2021;11(1):24477.

J1 **Murayama H**, Kayano T, Nishiura H. *Estimating COVID-19 cases infected with the Alpha variant (VOC 202012/01): an analysis of screening data in Tokyo, January–March 2021.* Theoretical Biology and Medical Modelling. 2021;18(1):13.

† Equal contribution.

Journal Correspondence

JC1 Jung S, Miura F, **Murayama H**, Funk S, Wallinga J, Lessler J, Endo A. *Preemptive Mpox Vaccine Deployment: Aligning Strategy with Reality.* The Journal of Infectious Diseases. 2025 Jul 21;jiaf365.

Governmental Reports

G2 Ko KY, **Murayama H**, Yamasaki L, Kinoshita R, Nishiura H, Suzuki M. *Evaluating the age-specific effectiveness of COVID-19 vaccines against death from surveillance data in Tokyo.* Materials 3-2, 65th Advisory Board Meeting, Ministry of Health, Labour and Welfare on COVID-19 Countermeasures, 28 Dec 2021. Link.

G1 Ko KY, **Murayama H**, Yamasaki L, Kinoshita R, Nishiura H, Suzuki M. *Evaluating the age-specific effectiveness of COVID-19 vaccines against death from surveillance data in Tokyo.* National Institute of Infectious Diseases, Infectious Diseases Surveillance Center. 2021 Dec. Link.

Conference Presentations

C2 **Murayama H**, Endo A. *Transmission dynamics and risk assessment of mpox clade IIb and Ib within men who have sex with men.* Early Career Researcher Sandbox session, Infectious Disease Modelling conference. Bangkok, Thailand. Oral presentation. Nov 2024.

C1 **Murayama H**. *Impacts of vaccine, healthcare burden, and temperature on the transmissibility or virulence of COVID-19.* COVID-19 Pandemic Conference. Nagoya, Japan. Oral presentation. Sep 2022.

Skills and Professional Development

Research areas: Infectious disease epidemiology and mathematical modelling of infectious diseases, including COVID-19, mpox clades I and II, sexually transmitted infections, and vaccine effectiveness; environmental epidemiology, focusing on interactions between infectious disease dynamics, temperature and tropical cyclones.

Programming: R, Julia, Stan.

Computing: JupyterLab, Docker, RStudio, Microsoft Office, GitHub, Zotero.

Markup: LaTeX, Markdown; familiar with HTML, XML, and CSS.

Methods: Bayesian methods, filtering methods, maximum likelihood estimation, differential equations, stochastic processes, branching processes, survival analysis, spatio-temporal modelling, network modelling.

Languages

Japanese: Native

English: Fluent; business level

Professional Services

Peer review: Co-reviewer for peer-reviewed journals: PLoS Neglected Tropical Diseases (1), PLoS ONE (1), The Journal of Infectious Diseases (1).